

Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY
PUNE-411037

Department of Engineering, Sciences and Humanities

Calculus and Statistics HS1076
UNIT II - FUNCTIONS OF SEVERAL VARIABLES
Tutorial and Assignment

1. If $u = \log(x^2 + y^2) + \tan^{-1}\left(\frac{y}{x}\right)$ then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.
2. If $u = \log(e^x + e^y)$ then show that $\frac{\partial^2 u}{\partial x^2} \cdot \frac{\partial^2 u}{\partial y^2} = \left(\frac{\partial^2 u}{\partial x \partial y}\right)^2$.
3. If $z^3 - zx - y = 0$ then find $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ and show that $\frac{\partial^2 z}{\partial x \partial y} = -\frac{(3z^2 + x)}{(3z^2 - x)^3}$.
4. If $u = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$ then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$.
5. If $z = \tan^{-1}\left(\frac{2xy}{x^2 - y^2}\right)$ then show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$.
6. If $z = \tan(y + ax) - (y - ax)^{\frac{3}{2}}$ then show that $\frac{\partial^2 z}{\partial x^2} = a^2 \frac{\partial^2 z}{\partial y^2}$.
7. Find the value of n if $u = Ae^{-gx} \sin(nt - gx)$ satisfies the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ where g and A are constants.
8. Find the value of n for which $u = t^n e^{-\frac{x^2}{4t}}$ satisfies the partial differential equation $\frac{\partial u}{\partial t} = \frac{1}{x^2} \left[\frac{\partial}{\partial x} \left(x^2 \frac{\partial u}{\partial x} \right) \right]$
9. If $z = f(x + ct) + \phi(x - ct)$ prove that $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$
10. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x + y + z)^2}$

1. Verify Euler's Theorem for (i) $u = \frac{x^2 + y^2}{x + y}$ (ii) $u = 3x^2yz + 5xy^2z + 4z^4$

2. If $u = \cos^{-1} \left[\frac{x + y}{\sqrt{x} + \sqrt{y}} \right]$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2} \cot u$.

3. If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$ then show that

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = (1 - 4\sin^2 u) \sin 2u$$

4. If $u = \operatorname{cosec}^{-1} \left[\sqrt{\frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}} \right]$ then prove that

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144} (13 + \tan^2 u).$$

5. If $u = \sin^{-1}(x^3 + y^3)^{2/5}$ then show that

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{6}{5} \tan u \left(\frac{6}{5} \sec^2 u - 1 \right)$$

6. If $z = f(u, v)$ and $u = x^2 - 2xy - y^2, v = y$, prove that

$$(x + y) \frac{\partial z}{\partial x} + (x - y) \frac{\partial z}{\partial y} = (x - y) \frac{\partial z}{\partial v}$$

7. If $z = f(u, v)$ and $u = x \cos \theta - y \sin \theta, v = x \sin \theta + y \cos \theta$, prove that

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v}$$

8. If $z = f(x, y), x = u + v$ and $y = uv$, prove that

$$i) (u - v) \frac{\partial z}{\partial x} = u \frac{\partial z}{\partial u} - v \frac{\partial z}{\partial v}$$

$$ii) (u - v) \frac{\partial z}{\partial y} = \frac{\partial z}{\partial v} - \frac{\partial z}{\partial u}$$

$$iii) x \frac{\partial z}{\partial x} + 2y \frac{\partial z}{\partial y} = u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v}$$

9. If $u = f(x^2 - y^2, y^2 - z^2, z^2 - x^2)$, then show that $\frac{1}{x} \frac{\partial u}{\partial x} + \frac{1}{y} \frac{\partial u}{\partial y} + \frac{1}{z} \frac{\partial u}{\partial z} = 0$.

10. If $u = f(e^{x-y}, e^{y-z}, e^{z-x})$, then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.

Tutorial and Assignment

Q. 1) Find extreme values of the following functions.

1.	$f(x, y) = 3x^2 - y^2 + x^3$
2.	$f(x, y) = xy + a^3 \left(\frac{1}{x} + \frac{1}{y} \right)$
3.	$f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$
4.	$f(x, y) = x^3 y^2 (1 - x - y)$
5.	$f(x, y) = 2(x^2 - y^2) - x^4 + y^4$
6.	$f(x, y) = x^3 + 3x^2 + y^2 + 4xy.$
7.	$f(x, y) = \sin x + \sin y + \sin(x + y)$
8.	$f(x, y) = xy(a - x - y)$ where $a > 0$
9.	$f(x, y) = x^4 + y^4 - 4xy + 1$
10.	$f(x, y) = x^2 + y^2 - 4y + 9$

Q. 2) Find the minimum distance from origin to plane $3x + 2y + z = 12$.

Q. 3) Divide 24 into three parts such that the continued of the first, square of the second and cube of the third may be maximum.

Q. 4) Show that if the perimeter of triangle is constant then triangle has maximum area when it is equilateral.

Q. 5) Show that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.

Q. 6) The temperature at any point (x, y, z) in space is $T = 400xyz^2$.

Find the highest temperature on surface of the sphere $x^2 + y^2 + z^2 = 1$.

Q. 7) The resonant frequency in a series electrical circuit is given by $f = \frac{1}{2\pi\sqrt{LC}}$.

If the measurement of L and C are in error by 2% and -1% respectively, find the % error in f .

Q. 8) The area of ΔABC is calculated using the formula $\Delta = \frac{1}{2}ab \sin C$.

Errors 2%, 3%, 4% are made in measuring a, b, C respectively.

If the correct value of C is 30° , find % error in the value of Δ .

Q. 9) Prove that $JJ' = 1$ for the following functions.

a) $u = e^x \sec y$, $v = e^x \tan y$

b) $u = x(1 - y)$, $v = xy$

Q.10) Verify whether the following functions are functionally dependent and if so, find the relation between (among) them.

a) $u = \frac{x - y}{x + y}$, $v = \frac{x + y}{x}$

b) $u = \frac{x^2 + y^2}{x}$, $v = \frac{y^2}{x}$